INCH-POUND

MIL-I-24768/21
25 February 1992

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, PAPER-BASE, PHENOLIC-RESIN (PBG-P)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a paper-base filler impregnated and bonded with a phenolic-resin compound or binder and processed to conform to this specification.

Intended use: The laminated material covered by this specification is moisture and humidity resistant, suitable for hot punching in electrical applications.

Forms: Sheets are the only recommended form of laminated material covered by this specification.

Temperature indice: The laminated material covered by this specification has a mechanical strength test index of 130 and an electrical strength test index of 130 for thickness up to 0.062 inch and a mechanical strength test index of 140 and an electrical strength test index of 140 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/21-S-1	0.010	0.002
M24768/21-S-2	0.015	0.0025
M24768/21-S-3	0.020	0.003
M24768/21-S-4	0.025	0.0035
M24768/21-S-5	0.031	0.0035
M24768/21-S-6	0.047	0.0045
M24768/21-S-7	0.062	0.005
M24768/21-S-8	0.094	0.007
M24768/21-S-9	0.125	0.008
M24768/21-S-10	0.156	0.009
M24768/21-S-11	0.188	0.010
M24768/21-S-12	0.219	0.011
M24768/21-S-13	0.250	0.012
M24768/21-S-14	0.312	0.014
M24768/21-S-15	0.375	0.017
M24768/21-S-16	0.438	0.019
M24768/21-S-17	0.500	0.021
M24768/21-S-18	0.625	0.024
M24768/21-S-19	0.750	0.027
M24768/21-S-20	0.875	0.030
M24768/21-S-21	1.000	0.033
M24768/21-S-22	1.125	0.035
M24768/21-S-23	1.250	0.037
M24768/21-S-24	1.375	0.039
M24768/21-S-25	1.500	0.041
M24768/21-S-26	1.625	0.043
M24768/21-S-27	1.750	0.045
M24768/21-S-28	1.875	0.047
M24768/21-S-29	2.000	0.049
M24768/21-S-30	2.125	0.052
M24768/21-S-31	2.500	0.056
M24768/21-S-32	2.750	0.061
M24768/21-S-33	3.000	0.065
M24768/21-S-34	3.500	0.073
M24768/21-S-35	4.000	0.082
M24768/21-S-36	4.500	0.090
M24768/21-S-37	5.000	0.095
M24768/21-S-38	5.500	0.105
M24768/21-S-39	6.000	0.115
M24768/21-S-40	6.500	0.120
M24768/21-S-41	7.000	0.130
M24768/21-S-42	7.500	0.140
M24768/21-S-43	8.000	0.145
M24768/21-S-44	8.500	0.155
M24768/21-S-45	9.000	0.160
M24768/21-S-46	9.500	0.170
M24768/21-S-47	10.000	0.180
M24768/21-S-48	_[1]	_[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet is: M24768/21-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Performance requirements: Performance requirements for laminated materials shall be as specified in table II.

The following performance requirements do not apply to this plastic laminate:

a. Surface resistance
b. Volume resistivity
c. Silicone content
d. Arc resistance
e. Thermal endurance
f. Flammability
g. Tracking resistance
h. Impact strength.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE II. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage	A D-48/50	4.6.4	Min. kV	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	60.0 5.0	3.0
Dielectric constant	A D-24/23 D-48/50	4.6.6	Max.	5.00 5.50	5.00 5.30	5.00 5.30	5.00 5.20 5.80	5.00 5.20	5.00 5.20				
Dissipation factor	A D-24/23 D-48/50	4.6.6	Max.	0.040 0.080	0.040 0.060	0.040 0.060	0.040 0.050 0.100	0.040 0.050	0.040 0.050				
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft-lbs. per inch										
Flexural strength Lengthwise: Crosswise:	A	4.6.12	Min. psi	14,000 12,000	14,000 12,000	14,000 12,000	14,000 12,000	14,000 12,000	14,000 12,000				

See footnotes at end of table.

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TABLE II. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength	A D-48/50	4.6.15	Min. pounds										
Water absorption	D_1-24/23	4.6.17	Max. percent	2.80	1.80	1.40	1.10	0.85	0.65				
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review Activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User Activities:
Army – ME
Navy – MC, AS

Preparing Activity:
Navy – SH
(Project 5970-1015-21)

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